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ORIGINAL RESEARCH

Fairness–accuracy tradeoff: activation function choice in a neural network

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Abstract
Models can have different outcomes based on the different types of inputs; the training data used to build the model will change its output (a data-centric outcome), while the hyperparameter selection can also affect the model's output and performance (a model-centric outcome). When building classification models, data scientists generally focus on model performance as the key metric, often focusing on accuracy. However, other metrics should be considered during model development that assesses performance in other ways, like fairness. Assessing the fairness during the model development process is often overlooked but should be considered as part of a model's full assessment before deployment. This research investigates the *fairness–accuracy tradeoff* that occurred by changing only one hyperparameter in a neural network for binary classification to assess how this single change in the hidden layer can alter the algorithm's accuracy and fairness. Neural networks were used for this assessment because of their wide usage across domains, limitations to current bias measurement methods, and the underlying challenges in their interpretability. Findings suggest that assessing accuracy and fairness during model development provides value while mitigating potential negative effects for users while reducing organizational risk. No particular activation function was found to be fairer than another. Notable differences in the fairness and accuracy measures could help developers deploy a model with high accuracy and robust fairness. Algorithm development should include a grid search for hyperparameter optimization that includes fairness along with performance measures, like accuracy. While the actual choices for hyperparameters may depend on the business context and dataset considered, an optimal development process should use both fairness and model performance metrics.

Keywords Fair · Algorithmic bias · Neural network · Bias · Machine learning · Deep learning · Society · Ethical frameworks

1 Introduction

Algorithms are becoming ubiquitous and many organizations have difficulty balancing the development and implementation with the risk that comes from an unfair or biased model. This research focused on how the selection of one hyperparameter—the activation function hyperparameter within a neural network—can affect both the accuracy and the fairness of the model.

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Accuracy is often considered the most important metric when building and deploying classification models. However, this performance-driven focus is shortsighted given that there is considerable risk when an unfair model is deployed. To mitigate risk, organizations must consider other aspects that affect the users of the algorithm, namely fairness.

While data scientists and organizations generally want to make accurate and fair models, fairness has been considered difficult to measure. Generally, neither internal organizational key performance indicators (KPIs) nor external governmental regulations require fairness or bias assessments.

There are many different definitions and subjective assessments of fairness. To establish a robust yet simple metric to assess fairness, this research used the Uniform Guidelines on Employment Selection Procedures [29 C.F.R. × 1607.4(D)] as outlined by the United States government [1]. This metric, often called “the 4/5 rule,” enables a comparison of the selection rates by group to assess fairness. The selection rate for a particular group by an algorithm is directly compared against selection rates for other groups to evaluate the overall fairness of the model.

For example, if the model’s selection rate for one group was 30%, and the selection rate for another group was 20%, the lower rate is divided by the higher selection rate to assess the *adverse impact* (20/30 = 66.6%); this is sometimes called a *disparate impact assessment*. For regulatory requirements in the United States, the threshold is 4/5 or 80%, meaning that the adverse impact (i.e., disparate impact) should enable ratios above 80%, thereby indicating that the groups have a similar selection rate. There are many critiques of this 4/5 threshold (it is just a *rule of thumb* and not intended to resolve unlawful discrimination), and this threshold will not be utilized within this research; however, the methodology of comparing the selection race between groups will be the measure utilized for fairness throughout this research.

1.1 Bias as the potential for unfairness

Algorithmic bias can lead to unfairness and is one of the most extensively discussed areas in the machine learning world. Bias caused by algorithms is perceived to be harmful, including parole recommendations [2], facial recognition systems [3], credit card or lending applications [4] emerged as prominent examples [5]. AI, algorithms, and autonomous systems can affect individuals and groups unfairly based on race, gender, or religion as well as other protected category variables. Bias is harmful when (a) it contributes to unfair discrimination, (b) which is systemic and (c) leads to unfair outcomes. Bias not only results in discrimination, but also impacts the trust of the users of such system Fairness is one of the components needed for trustworthy AI [6].

1.2 Categories of bias

Bias in computer systems can be categorized as pre-existing, technical, and emergent bias [7]:

- Pre-existing bias is contributed by an individual or societal biases in the environment. These are also amplified by the selection and representation of the data gathered. Much of the discussions around bias lead to data bias or the underlying societal bias.
- Technical bias can arise from tools, lack of context for the model, and inconsistencies in coding abstract human concepts into machine learning models.
- Emergent bias arises from the feedback loop between human and computer systems.

The existing studies mainly discuss bias, unfairness, and discrimination in general but rarely delve into detail by studying what kind of bias occurred [8] or ways to mitigate it. Algorithms trained on biased data are perceived to contribute to inequalities in outcomes, thereby putting certain groups at a disadvantage [9].

Some researchers assessed how hyperparameter optimization should be conducted and reported, while [10] others reviewed how metrics used to assess deep learning neural networks can cause suboptimal model adoption [11], but no researchers have assessed how the selection of hyperparameters can affect the technical bias in deep learning and thereby the fairness to users of the model.

This paper argues that technical contributors within a model (i.e., hyperparameter selection) will affect accuracy and fairness, and steps should be taken to assess both as a means to deploy highly accurate models that are also the fairest models.

1.3 Experimental study

This research approaches the fairness-accuracy tradeoff by assessing how one small change in the model hyperparameters affects the overall accuracy and fairness between groups.

The systematic changing of the hyperparameters for each model inserts a small amount of technical bias; model accuracy and fairness are assessed to determine the magnitude of the change.

The paper is also expected to contribute to “*exposing the knobs of responsibility to data scientists*,” [12], to identify the potential for technical bias introduced into modeling with each choice made by the data scientist during model development. While no particular hyperparameter is expected to be fairer than another, we do expect the methodology of assessing the fairness and the accuracy of the model is one of the process “knobs” every data scientist needs to wield.

The contribution of this research is the novel concept that a model’s fairness can be reduced or improved by hyperparameter choice in artificial neural networks (ANN).

The experimental study examined the relative impact of model component hyperparameter choices on a protected group. We choose a multi-layer ANN for testing model component hyperparameters within it because of its wide applications across domains, limitations to current bias measurement methods, and the underlying challenges in interpretability.

An ANN is a collection of nodes that act as artificial neurons that attempt to loosely mimic the behavior of the biological brain [13]. The advantages of using an ANN are its non-linearity and its ability to learn without being reprogrammed. This research focuses on an ANN with tabular data.

ANNs are applied in several use cases, including recommendation engines, anomaly detection, medical diagnosis, facial recognition, college admissions, and credit rating. Neural networks, as a whole, are often criticized for their *black box* nature [14] and [15]. The term "black box" generally applies to a model in which a data scientist can see inputs and outputs but has difficulty fully interpreting the model; this is often called the explainability problem of black box models. ANNs also suffer from explainability: it is difficult to explain why a certain result was obtained. Even small perturbations of input data (i.e., training data) can have a considerable effect on the output leading to completely different results [16]. This is of great importance in virtually all critical domains where we suffer from poor data quality. Even with the existing methods, the interpretation of ANNs may be quite divergent and may require domain knowledge to reason [17].

Assessing the fairness in neural networks requires assessing performance and outcomes for specific groups. Hence, the awareness of the change in fairness contributed by the model hyperparameter can be helpful in dissecting the interpretation and bias measurement methods. These aspects provide the reasoning for examining the neural network component hyperparameter contribution to bias, thereby furthering the drive for fairness.

1.4 Neural network and parametric choices

While there are many hyperparameters requiring optimization within a neutral network (e.g., dropout rate, number of layers, nodes per layer), neural networks contain three key model hyperparameters, namely the activation function, optimizer, and loss function. These components are used across layers for constructing the neural network. This research focused on the fairness changes of the chosen activation function by conducting an experimental study on two separate data sets.

In this experimental study, we run the ANN on each dataset by adjusting only the activation function hyperparameter in the hidden layers of the neural network and holding all other aspects constant, thereby making the activation function hyperparameter choice the only impact (i.e., variation) between on the model. The model's performance accuracy and fairness were assessed using the open-source toolkit Fairlearn [18].

1.5 Motivation

One of the main motivations for the research was to find a way to operationalize the assessment of fairness during model development and the recognition that choices that developers make have an impact on the accuracy and fairness of the model. For example, the ArgMax activation function could amplify bias within a neural network, but developers would never know because fairness is not typically measured.

This research focuses only on the activation functions in the hidden layers to see if they "may amplify tiny data biases into perfectly biased outputs" [19] and affect the fairness among groups.

The paper covers the prior research on the subject in Sect. 2, the experimental research methodology in Sect. 3, research outcomes and their analysis in Sect. 4, broad-level recommendations and a conclusion in Sect. 5.

2 Prior research

Often, the focus on bias is toward the bias in data, instead of addressing more subtle forms of bias [20]. Machine learning courses often focus on the optimization of the neural network without focusing on the potential pre-existing, technical, or emergent biases; minimizing computational time and cost while training a highly accurate model is often the main consideration of algorithm developers.

Kate Crawford, in her NEUR IPS 2017 keynote, described biases that cause allocative harm as "representational harm" (i.e., harmful representation of human identity), thereby exhibiting a need to extend beyond data bias [21]. To fully understand the impact of models on people, it is necessary to understand the facets of technical bias.

Other researchers found measuring fairness and reporting fairness metrics to users (i.e., the people actually interfacing with the algorithm) improved their trust of the artificial intelligence (AI) model [22]. Other research also analyzed the impact of fairness on trust; there is a synergistic effect of how fairness affects users' trust in AI and machine learning (ML) models [5].

2.1 Understanding roots of bias

Technical bias usually emerges within the design process of the algorithms and is the result of limitations of computer tools, such as hardware and software [23]. Inconsistencies in codifying abstract human concepts include attempts to quantify the qualitative and discretize the continuous [24]. Technical bias can also appear if the model becomes oversimplified because of constraints, thereby creating or contributing to bias [25]. Most discussions are about pre-existing bias and its impact on technical efforts, including efforts to debias the data.

Technical bias is described as "computer tools bias, decontextualized algorithm bias, random number generation and codifying abstract human concepts" [7] and also described as measurement bias, modeling bias, label bias (representation impact), and optimization bias (tradeoff impact) [26].

Technical bias is represented as an epistemological problem [27] that requires justification and value judgment. We extend that argument to select aspects of technical bias, specifically, the choice of model hyperparameters.

In one of the recent papers, the authors examine the bias contributed by pre-processing and post-deployment issues [12]. This is one of the few articles that examines technical bias. Studies have discussed inconsistent results contributed by choices made in model hyperparameters [10] in research, requiring researchers to disclose the methods adopted for hyperparameter optimization. However, the independent impact of how model hyperparameters have a role to play in bias amplification and how parametric choices are made has not been studied.

3 Research methodology

3.1 Data

Two datasets were utilized to investigate the potential technical bias from hyperparameter selection. Both datasets are open and available and utilize people as the unit of analysis. Both datasets are open-source and used by other researchers [27] and [28]. (Appendix A).

3.1.1 French E-commerce data

This dataset "was scraped from a successful online C2C fashion store with over 9 M registered users" [29]. The dataset has 98,913 records, each record representing an online customer. For explainability purposes, data pre-processing was conducted in Alteryx Designer and included outlier handling; high outliers were identified as numerical values greater than three standard deviations from the mean and the whole record was removed ($n = 375$, 87 male and 287 female). Outlier handling yielded a final dataset of 98,538 records.

Categorical variables were one hot-encoded, and numerical values were scaled to standardize values from 0 to 1. To enable a binary classification, the binary variable "transact" was made to bin customers into two groups: "1" if they made a transaction (bought or sold an item on the online retailer) or "0" if there were no transactions. After pre-processing was complete, there were 18 independent variables and one target variable: *transact*.

The stratified random sample was based on the gender variable to maintain the same ratio of males to females in the training and testing data because the French e-commerce data was more unbalanced with regard to the protected group variable, *gender*, with 76.6 of the customers identifying as female. The 70/30 data split made a training dataset of 68,976 records (15,893 male; 53,083 female) and a test dataset of 29,562 records (6812 male; 22,750 female). An unstratified split could have impacted the prediction and/or bias metrics and made the experimental nature of the analysis less controlled. A static seed was used in the data split for replicability.

The research and evaluation of fairness were made toward the specific reference to *gender* as a protected category variable. The preprocessed data was exported from Alteryx Designer as a CSV file and imported into Google Colab Python IDE to use the Keras library for modeling.

3.1.2 Adult

This dataset is at the individual level (i.e., each record is a person) and from the 1994 Census [30]. The dataset is widely used with over 2.5 million downloads and contains "gender" as a key protected category variable. The dataset (Dataset 2) had a binary target variable, "Income > = $50,000" where "1" represented a person who makes over $50,000 a year and "0" was an adult who did not make that much. The dataset is well suited for experimental research because it was balanced between the male and female gender groups (i.e., a protected category variable) and required reduced data pre-processing compared to the dataset (i.e., Dataset 1).

Dataset 2 had 45,222 records with each record representing an adult. The data already came in separate training and testing files; as part of pre-processing, the testing and training data files were merged to make the normalization steps standardized. A wholly separate stratified random sampling was conducted of the merged data.

Records with null values on total charges were removed from the analysis. In addition, one hot encoding was performed to convert each categorical variable to several dummy variables. The dataset included a continuous variable and a categorical variable for education attainment; the categorical education variable was removed from the analysis because it introduced 16 variables once encoded thereby reducing the potential for multicollinearity. Continuous variable values were scaled to standardize values from 0 to 1. The gender variable was converted to "1" and "0" for "Female" and "Male," respectively. For modeling, the final dataset had 87 independent variables and one target variable: *Income > $50 k*.

Like the first dataset, the stratified random samples were based on the *gender* variable and the resulting 70/30 data split yielded a training dataset of 31,656 records (21,369 male; 10,287 female) and a test dataset of 13,566 records (9,158 male; 4,408 female). A static seed was again used in the data split for replicability.

Table 1 Model variations for each experiment

Experiment	Notation	First hidden layer	Hidden layer variation	Output layer
1	ReLU-ReLU-Sigmoid (RRS)	ReLU[a]	ReLU	Sigmoid[b]
2	ReLU-Tanh-Sigmoid (RTS)	ReLU[a]	Tanh	Sigmoid[b]
3	ReLU-ELU-Sigmoid (RES)	ReLU[a]	ELU	Sigmoid[b]
4	ReLU-SELU-Sigmoid (RSS)	ReLU[a]	SELU	Sigmoid[b]

[a]The activation function for the input layer was held constant. Only the activation function for the Hidden Layer changed.

[b]The experiments are binary classifiers, therefore the experiments share the Sigmoid activation function for the output layer

3.2 Neural network experiments

The experiment utilized a binary classification model using a neural network with 3 layers: 2 hidden layers, and 1 output layer—on Google Colab Python Notebook using the Keras library. Building a neural network is an empirical process, so it is important to control all possible hyperparameter settings. The sigmoid activation function was considered as the static activation function in the output layer aligning with the common practice for binary classification. (The code repository is available in Appendix B.)

Four experiments were conducted utilizing the two different stratified samples for each dataset (Table 1):

Experiment Model 1: Rectified linear unit (ReLU) was the default hidden layer activation function in many neural networks [31] and [32]. Hence, we considered the ReLU in the input and hidden layer, and sigmoid in the output layer for our Baseline Model 1. The activation function sequence for the NN would be

Experiment Model 2: The activation function sequence for the ANN would be the ReLu as the activation functions in the first hidden layer –> Hyperbolic tangent (Tanh) activation functions in the second hidden layer –> sigmoid for the output layer.

Experiment Model 3: The activation function sequence for the ANN would be the ReLu as the activation functions in the first hidden layer –> Exponential linear unit (ELU) activation functions in the second hidden layer –> sigmoid for the output layer.

Experiment Model 4: The activation function sequence for the ANN would be the ReLu as the activation functions in the first hidden layer –> Scaled exponential linear unit (SELU) activation functions in the second hidden layer –> sigmoid for the output layer.

The experiment changed the activation function in the hidden layers keeping everything else in the baseline model constant. The activation functions we considered for the second hidden layer for the experiments were the exponential linear unit (ELU), rectified linear unit (ReLU), exponential, scaled exponential linear Unit (SELU), and hyperbolic tangent (Tanh). The output layer was retained as Sigmoid as a standard activation function for the binary classification.

The Binary Cross Entropy Loss Function and Adam Optimizer were utilized as constants for all the experiments. Binary Cross Entropy is the recommended loss function for binary classification in Keras.

3.3 Libraries used

The key libraries we used for the purpose of the research include Tensorflow, Keras, NumPy, Sklearn, and Fairlearn (Appendix B).

3.4 Activation functions

Four activation functions were selected from the nine options available in the Keras Python library.

Each of the four experimental activation functions was selected due to their broad applicability. Each activation function has different outcome ranges with the ELU and SELU being the most similar. All activation functions, specifically ELU and SELU, utilized Keras defaults (Table 2).

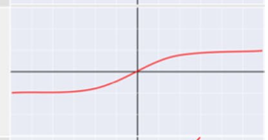
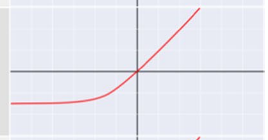
3.5 Performance metrics

We gathered the key performance metrics of each experiment for the protected category variable (gender) and populated them alongside the performance of the baseline model. The performance metrics considered for this purpose were focused on accuracy; precision and recall could also be considered but were excluded from the analysis. In addition, fairness-specific metrics were considered using Fairlearn, as described below.

3.6 Fairlearn

Fairlearn was the primary open-source Python library used to assess the fairness in the neural networks. Fairlearn assesses fairness from the perspective of two kinds

Table 2 Activation functions overview

Activation Function	Equation	Range	Plot
ReLU	$\mathrm{relu}(z) = \begin{Bmatrix} z & z > 0 \\ 0 & z <= 0 \end{Bmatrix}$	(0, ∞)	
TanH	$\tanh(z) = \dfrac{e^{z} - e^{-z}}{e^{z} - e^{-z}}$	(-1, 1)	
ELU	$\mathrm{elu}(z) = \begin{Bmatrix} \alpha(e^{z} - 1) & z < 0 \\ z & z >= 0 \end{Bmatrix}$ parameter α, default =1	(-α, ∞)	
SELU	$\mathrm{selu}(z) = \begin{Bmatrix} \lambda\alpha(e^{z} - 1) & z < 0 \\ \lambda z & z >= 0 \end{Bmatrix}$ parameters α ≈ 1.67326 λ ≈ 1.05070	(-λα, ∞)	

of harms, namely, (a) Allocation harms and (b) Quality of Service Harms based on the group fairness approach. Fairness metrics covered in Fairlearn were false positive rate, false negative rate, true positive rate, true negative rate, and selection rate.

To align Fairlearn's analysis with the Equal Employment Opportunity Commission (EEOC) guidelines, the method to identify *disparate impact* was utilized. Disparate impact, also called *adverse impact*, assesses the differences in the selection rate of groups.

It is common to determine the algorithm as unfair if the ratio of the selection rates for the groups is less than 4/5th of the group with the highest rate, then this scenario is considered an *adverse impact* (has a *discriminatory effect*) on a group [1]. However, the goal of this experiment is to assess how changing the model's hyperparameters affects fairness, so the method of measuring adverse impact is a proxy for fairness, therefore the 4/5th threshold is not used. A model's fairness metric is only compared to other iterations of the same model on the same dataset; for this research, the four models—each with a different activation function in the hidden layers—are compared to themselves. The fairness of models built with different datasets is not directly compared.

3.7 Hyperparameter standardization

We anticipated the potential possibility of hyperparameters impacting the prediction accuracy and the fairness metrics because ANN's are non-deterministic due to the random initialization of weights and biases. To mitigate side effects from the randomization process during ANN initialization, certain hyperparameters were standardized for this experiment. The hyperparameters that we standardized were the number of layers, nodes for each layer, random seeds, learning rate, dropouts, batch size, and epochs. The selection of most hyperparameters was not for optimization but for standardization within the code to test for bias influence between activation functions.

The ANN architecture and the number of layers and nodes are controlled for each dataset because the number of layers and size of the neural network will have an impact on model training and hence, we structured the ANN to be standardized for all the experiments. The structures of the ANNs for both datasets matched except for the input layer and the first layer. For the French E-Commerce Data (dataset 1), the ANN architecture was 19 input nodes, 27 nodes in the first hidden layer, 5 nodes in the second hidden layer, and 1 node in the output layer (19, 27, 5, 1). The overall ANN architecture for the Adult Census dataset (dataset 2) had a similar structure in every way to the first dataset, except the input layer was adjusted to match the dataset's input features: 87 input features so 87 nodes for the input layer, 27 nodes of the first hidden layer, 5 nodes of the second hidden layer, and 1 node in the final layer (87, 87, 5, 1).

Seed: Assigning a set seed for the Python packages enabled reproducible results over many iterations. There are 3 requirements for seed in our experimental model, namely, (1) Tensorflow random seed, (2) NumPy seed, and (3) Keras initializer seed. The seeds were standardized to avoid potential inconsistencies in the outcomes from neural networks and thereby an impact on fairness metrics measured to study the bias contributed by activation function [33] and [34].

Learning rate: The learning rate is one of the configurable hyperparameters in a neural network, which determines the rate at which the model learns in response to model weight updates each time. Larger learning rates result in

unstable training and tiny ones fail to train. The learning rate was set by the Adam optimizer in Keras at 0.001, to eliminate possible bias contributed by it in our study [35].
Dropouts: Dropout involves the process of reviewing the nodes in each of the layers and determining the probability of retaining each node. Such a process ensures that it reduces the potential of overfitting the model. To ensure replicability, there are no dropouts in the models used for this experiment [36] and [37].
Batch Size: Batch size is the number of samples passed through the network at a time. Batch sizes are normally powers of two (e.g., 32, 64, 128, and 256). The batch size was held constant throughout the experiment at 256 because changes in batch size may have an impact on the performance of neural networks [38] and [39].
Epochs: Epochs determine the number of passes of the training dataset the model has completed. As the epochs increase, the model will help in better accuracy, before progressing toward overfitting. Epochs are determined based on an iterative study of optimal model outcomes. We have considered 30 epochs for all experiments of our neural network because the models converged prior to 30.

All other hyperparameters that are unmentioned are held constant to showcase the difference between the models based only on the changes to the activation function in the two hidden layers.

4 Experiment outcomes and analysis

4.1 Experiment outcomes

The experiments provided the following key outcomes in Table 3.

The accuracy rates for the four activation functions are very consistent with a range of 0.0009 and a standard deviation of 0.0003. The activation function seems to have little effect on the accuracy. However, the fairness measure varied considerably, with a range of 0.1309 and a standard deviation of 0.0471. If accuracy alone was considered, Selu has the highest accuracy score. If Fairness is also considered, the Relu model is nearly as accurate and has a considerably higher fairness score in Table 4.

The accuracy rates for the four models built with the Adult dataset (Table 4) vary more than the models built with the French data (Table 3). The range is ten times larger at 0.0914 and the standard deviation is 0.0379. The fairness measure varied less, with a range of 0.0567 and a standard deviation of 0.0204. If accuracy alone was considered, Selu has the highest accuracy score. If Fairness is also considered, then the Selu is also the best. While the most accurate model also had the highest fairness score, we would not have identified this without assessing both.

4.2 Analysis

Concerns around fairness have grown alongside the expansion of neural network use. For instance, neural networks used in facial recognition are found to be biased against people of color [15]. These are unintended biases introduced or amplified by the neural networks. One of the prominent root causes for such unintended bias is the pre-existing bias in the training data. However, there are other causes. This paper establishes that choice of model hyperparameters (in this case, the neural network's activation function in a hidden layer) can also contribute to biased outcomes for protected category variables (gender). Analysis of metrics and understanding of potential bias impact from the experiment provide the following insights:

The experiments in this research show that the choice of activation function in hidden layers might have an impact on fairness outcomes for the two datasets analyzed and accuracy, holding all other aspects constant.
Developers often choose activation functions to minimize processing time (e.g., ReLU is considered to be time efficient in training neural network models [40]) or to optimize a performance metric (e.g., accuracy or RMSE); however, there is a need to examine appropriate ethical choices between efficiency in processing and outcome fairness.
While the experiments suggest that the choice of activation function contributes to bias (incremental or decre-

Table 3 French data (dataset 1)

Activation	Accuracy	Fairness[a]
Elu	0.9425	0.8535
Relu	0.9422	0.9844
Selu	0.9429	0.9351
Tanh	0.9420	0.9368

[a]Fairness is the selection rate of males divided by the selection rate of females

Table 4 Adult data (dataset 2)

Activation	Accuracy	Fairness[a]
Elu	0.8435	0.2944
Relu	0.8368	0.2895
Selu	0.8378	0.3277
Tanh	0.7521	0.2710

[a]Fairness is the selection rate of males divided by the selection rate of females

mental), it does not establish that one activation function is fairer than another.

There exists no comprehensive solution to fairness in neural network modeling. While the research establishes the bias influence of the activation function, it does not establish if one activation function is fairer than the other. Studies have often critiqued the choices made on hyperparameters are not always supported by justifications [9].

5 Recommendation and conclusion

5.1 Recommendation

With the assessed influence of the activation function on fairness and accuracy, we believe data scientists need awareness of the existing bias contribution by different elements of the model to help make appropriate choices to mitigate unfairness.

In business settings, there is a common maxim that "*we can't fix what we can't measure*." In this situation, developers often assess the accuracy and fit but do not systematically assess how changes to hyperparameters or data affect model fairness. We propose data scientists build challenger models with different model hyperparameters and use a grid search approach to understand the metric disparities between different model hyperparameters to make appropriate choices of models that limit the introduction or amplification of bias. Performance metrics and fairness metrics should be used to assess the models.

Further, this approach helps arrive at a balance between efficient processing and fair user outcomes. This decision is an ethical choice made by a data scientist in designing and developing operational models in context. Said in a different way, the data scientist chooses to identify fair outcomes for different groups while striving for a highly accurate model that can be applied. In essence, fairness and efficiency need to be integrated into the approach besides model accuracy.

5.2 Conclusion

Fairness concerns have long been studied in the context of neural networks. However, specific studies on bias influence by model hyperparameter choices have not been studied extensively. This paper's primary contribution is to establish the fairness influence of model hyperparameter choice, specifically the activation function. This paper attempts to bring focus to the fairness-accuracy tradeoff and expands the understanding of the underlying contributors of technical bias in neural networks.

While the results do not modify the group fairness-based mitigation approaches adopted, we recommend the need for evaluating challenger models with varied model hyperparameters to make an ethical choice of the model that has (a) minimal bias influence and (b) optimal efficiency, given the fairness objective.

The paper has a couple of unresolved open questions. They are (a) underlying mathematical reason for variation in bias influence for different activation functions, (b) examining the coherence or divergence between efficiency and fairness caused by hyperparameter choices, and (c) understanding how the choice of hyperparameter may affect one or more protected category groups and their intersections (e.g., gender and race, black females). We believe these could become specific research questions soon. The paper does not recommend any preferred hyperparameter choices. Instead, it recommends that data scientists consider the fairness objective in the process of developing a neural network.

If there is such a difference between model outputs for accuracy and fairness based on changing one hyperparameter, there will be considerable differences when developers change data and hyperparameters.

Future work should include a full evaluation of the models using Shapley values and additional datasets to develop a deeper understanding of how hyperparameter optimization for accuracy affects fairness.

The nature of neural network modeling is a *black box* that is not necessarily understood even by experts, so data scientists should use an empirical process to assess how the structure of the model (i.e., hyperparameters) affects the bias on protected groups.

Appendix A

Data sources

1. Adult dataset
   Source: Dua, D. and Graff, C. (2019). UCI Machine Learning Repository. Available at: http://archive.ics.uci.edu/ml/datasets/Adult. Irvine, CA: University of California, School of Information and Computer Science [Accessed 01 Sept. 2022].
2. French dataset
   Source: Mabilama, J. M. (2020). E-commerce—Users of a C2C fashion store. [online] Data.world. Available at: https://data.world/jfreex/e-commerce-users-of-a-french-c2c-fashion-store [Accessed 05 Jan. 2022].

Appendix B

Code and Data Repository: https://github.com/Michael-AI-ML/fairness

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Data availability The data and code are made available in the Github listed in Appendix B and supplementary materials.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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